

A
Mini Project-II
Synopsis on
Smart Phishing Detection & URL Risk Analyzer

by

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Mini Project-II Synopsis

Mini Project-II Title: Smart Phishing Detection & URL Risk Analyzer

1. Problem Statement:

Phishing attacks are one of the most common cyber threats where attackers use fake or malicious websites to steal sensitive user information such as passwords, banking credentials, and personal data. Many users are unable to identify suspicious URLs, which increases the risk of financial loss and data breaches.

2. Objective:

- To develop a system that can analyze URLs and identify potentially malicious or phishing links.
- To provide users with a clear risk assessment of suspicious URLs before they access them.
- To improve user awareness and safety while browsing the internet.
- To assist in reducing the impact of phishing attacks by detecting suspicious patterns in web links.

3. Introduction:

The increasing use of internet services for communication, banking, education, and online transactions has also increased the number of cybersecurity threats faced by users. One of the most common threats is phishing, where attackers create deceptive websites or links that appear legitimate in order to trick users into revealing sensitive information such as passwords, financial details, or personal data. Many users access unknown links without verifying their authenticity, which increases the chances of falling victim to phishing attacks. A system that can analyze suspicious URLs and provide a clear risk indication can help users make safer decisions before visiting unknown websites. The Smart Phishing Detection & URL Risk Analyzer aims to evaluate web links and identify potential risks by examining different characteristics of the URL. By providing a simple and understandable risk assessment, the system can assist users in recognizing suspicious links and improving overall online safety.

4. Motivation:

Phishing attacks have become one of the most serious cybersecurity threats affecting individuals, organizations, and financial institutions worldwide. Attackers continuously create deceptive websites that look similar to legitimate platforms, making it difficult for users to distinguish between safe and malicious links. Existing security mechanisms often require advanced technical knowledge or specialized tools, which many users do not have access to. As a result, people frequently fall victim to phishing scams. Developing a system that can quickly evaluate the risk of a URL can significantly enhance user safety while browsing the internet. This project is motivated by the need to improve online security awareness and provide a simple tool that helps users detect suspicious links before interacting with them. Such systems have strong relevance in cybersecurity, digital safety, and internet security research.

5. Existing System:

Currently, several systems and tools are used to detect phishing websites and malicious URLs. These include browser security features, online threat intelligence services, and antivirus solutions that attempt to identify suspicious domains. Many web browsers also include basic security warnings that alert users when they attempt to access known malicious websites. However, these systems often rely mainly on previously reported phishing databases or blacklists. If a malicious website is newly created or not yet listed in these databases, the system may fail to detect it. Some advanced detection systems require specialized cybersecurity tools or technical expertise, which may not be accessible to common users.

Drawbacks / Limitations / Challenges in Existing System:

- Many detection systems depend heavily on pre-existing threat databases.
- Newly created phishing websites may bypass traditional blacklist systems.
- Some tools are complex and not user-friendly for non-technical users.
- Limited awareness among users about identifying suspicious URLs.
- Lack of simple systems that provide quick risk evaluation for unknown links

6. Scope of the Project:

The scope of this project is to develop a web-based application capable of analyzing URLs and identifying potential phishing risks. The system will allow users to submit a URL and receive a risk evaluation based on various indicators.

The project focuses on identifying suspicious characteristics in URLs and presenting the results in a clear and understandable format. The system will help users determine whether a link may be safe or potentially malicious before visiting it.

The project includes functionalities such as URL analysis, risk evaluation, and result visualization. However, the system will be limited to analyzing web links and will not directly block websites or provide full network-level protection. The project is intended as a cybersecurity awareness and analysis tool rather than a complete security solution.

7. Methodology/Approach:

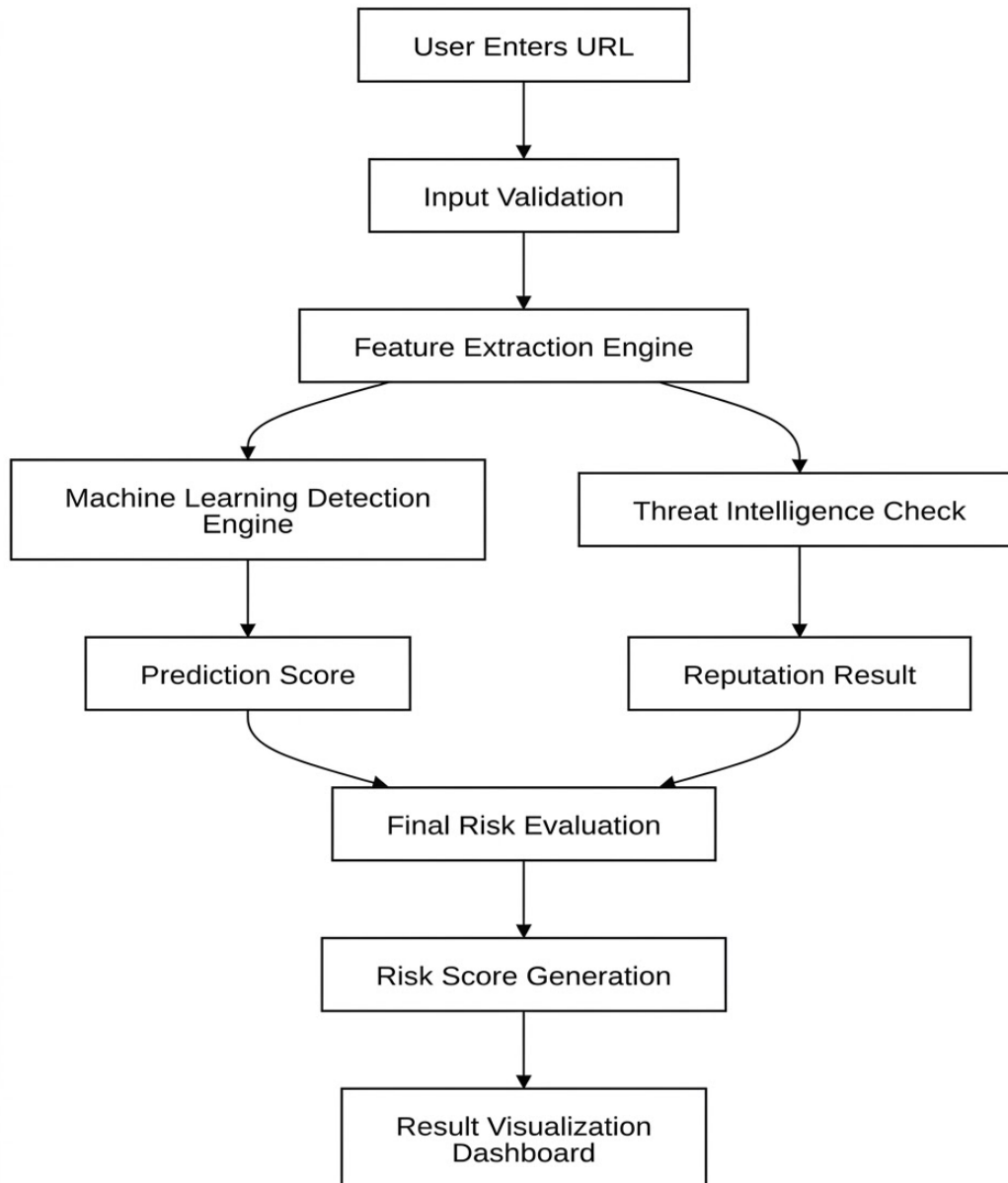
The project will follow a systematic development approach involving problem analysis, system design, implementation, testing, and evaluation. Initially, research will be conducted to understand phishing attack patterns and common characteristics of malicious URLs. Based on this analysis, a system will be designed to process and evaluate user-provided URLs. The system will analyze different properties of the input URL and determine a risk level. The final output will provide users with a clear indication of whether the URL appears safe or potentially malicious.

A. Tools and Technologies:

- Python
- React
- Tailwind CSS
- HTML
- JavaScript
- Git (Version Control)

B. System Design

Architecture Type: Client–Server Architecture

**C. Algorithms / Techniques:**

- URL pattern analysis
- Keyword-based detection techniques
- Risk scoring methods
- Basic heuristic analysis techniques

D. Modules

- User Interface Module
- URL Input Processing Module
- URL Analysis Module
- Risk Evaluation Module
- Result Display Module

E. Software Requirements

- Operating System: Windows / Linux
- Programming Language: Python
- Frontend Technologies: React, Tailwind CSS
- Development Tools: VS Code
- Browser: Google Chrome / Firefox

F. Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free space
- Internet Connection

8. References:

i. Journal/Conference Papers:

1. A. K. Jain and B. B. Gupta, “Phishing Detection: Analysis of Visual Similarity Based Approaches,” Journal of Information Security.
2. M. Aburrous, M. A. Hossain, K. Dahal, and F. Thabtah, “Intelligent Phishing Detection System for E-Banking Using Fuzzy Data Mining,” Expert Systems with Applications.
3. S. Verma and A. Das, “Phishing Detection System: Ensemble Approach Using CharacterLevel CNN,” arXiv preprint, 2025.

ii. Web References

1. <https://www.virustotal.com>
2. <https://phishtank.org>
3. <https://owasp.org>

iii. Book References

1. William Stallings – Network Security Essentials
2. Nina Godbole – Information Systems Security